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Review article

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Inheritance and selection of the mitochondrial genome

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Abstract

The functioning mitochondrial genome is essential for cellular energy production. Being strictly maternally inherited and possessing limited DNA repair capacity, mitochondrial DNA (mtDNA) replication errors tend to accumulate over time. If left unchecked, these errors can accumulate through the female germline over successive generations, potentially leading to species extinction. However, this outcome is not observed in most species, including humans, which implies the existence of mechanisms that counteract the progressive accumulation of deleterious mtDNA mutations. Recent technological advances are building a deeper understanding of the processes that preserve mtDNA integrity, including the molecular and cellular basis and timing of purifying selection. This new knowledge helps to explain how mtDNA can change rapidly over just a few generations, whilst remaining compatible with the independently inherited, evolving nuclear genome.

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Introduction

Mitochondria provide the major source of cellular energy, generating ATP through oxidative phosphorylation (OXPHOS)[1,2] and have important roles in other physiological processes[1–4], including calcium regulation and cell death[1–4]. Given their essential roles in cellular metabolism, disruption of mitochondrial DNA (mtDNA) integrity can affect a wide range of cell types, with particularly pronounced functional consequences in high-energy-dependent organs such as the heart, brain, eye and skeletal muscle[5–7]. mtDNA mutations cause rare inherited diseases, including Leber hereditary optic neuropathy, mitochondrial encephalomyopathy, lactic acidosis and stroke-like episodes (MELAS), and Leigh syndrome. They also contribute to common late-onset disorders, such as Alzheimer disease, cancer and type 2 diabetes mellitus[8,9].

Strict maternal inheritance[10–12] and the absence of meiotic recombination mean that deleterious mtDNA mutations cannot be efficiently eliminated by Mendelian inheritance[13,14] (Box 1). Hermann Muller recognized this vulnerable property of all uniparentally inherited genomes, including mtDNA, proposing a 'ratchet' mechanism by which mutations accumulate irreversibly across generations. Muller predicted that, if left unchecked, this process would eventually lead to species extinction[13,15] – a prediction supported by experimental and evolutionary studies in viruses[16], bacteria[17], yeast[18] and worms[15]. However, this outcome has not occurred in most species, including humans, which suggests the presence of mechanisms that prevent mtDNA 'mutational meltdown'. Indeed, in human populations, the frequency of pathogenic mutations at low levels of heteroplasmy (that is, mutations present in only a subset of mtDNA molecules) does not align with the incidence of mtDNA diseases, which is ~1 in 200 (refs. 19–21) versus ~1 in 5–10,000 (refs. 22,23), respectively. This observation implies that selective forces act against the transmission and accumulation of deleterious mtDNA variants.

As early as the 1980s, it was noted that most substitutions in mtDNA do not affect the amino acid sequence[24,25], a finding later validated in large-scale population studies[26–29]. In 2008, comparing the transmission of synonymous to non-synonymous variants across generations in mouse pedigrees provided strong evidence for purifying selection (also known as negative selection), in which deleterious variants are selectively removed during germline transmission[30]. Subsequent laboratory and population-based studies showed signatures of selection in various species, including *Caenorhabditis elegans*[31], *Drosophila melanogaster*[32–35], mouse[36–38] and human[39–42]. However, the underlying mechanisms remained challenging to study. The recent development of targeted mtDNA editors, such as the DddA-derived cytosine base editor (DdCBE)[43], has enabled the construction of precise, disease-specific mtDNA mutation models. Subsequent advances in oocyte in vitro culture, single-cell sequencing, imaging and analysis techniques have further accelerated experimental investigation, contributing to our understanding of when and how purifying selection maintains mitochondrial genome integrity across generations.

Here, we review the mechanisms influencing mtDNA inheritance, including both the timing and the underlying cellular and molecular processes involved. We focus on mammals, although we contrast mammal data with relevant invertebrate data, when available. We concentrate on inherited mtDNA variants rather than de novo mutations in the germline, for which mechanistic evidence remains limited despite their importance in the context of mtDNA disease origins and possibly fertility in otherwise healthy women[44,45]. Similarly, we do not consider somatic tissues, although we recognize that somatic mutational processes can indirectly influence the transmission of mtDNA variants (for example, through effects on organismal fitness or fecundity)[45,46], and we recognize emerging evidence that the underlying molecular mechanisms may be similar in somatic and germline contexts[47].

mtDNA inheritance and the mitochondrial bottleneck

In mammals, mtDNA is exclusively maternally transmitted[10–12] (Box 1); offspring inherit a population of mtDNA molecules that can include both wild-type and mutant genomes, a phenomenon known as heteroplasmy. The mutation rate of mtDNA is approximately 20 times higher than that of the nuclear genome, leading to the frequent occurrence of mutations[48,49]. Although most mtDNA variants remain at low heteroplasmy (<1%)[21,50], the proportion of mutant and wild-type mtDNA molecules can change markedly between generations (Fig. 1a). Some variants increase over successive generations and occasionally reach homoplasmy, the situation in which all mtDNA molecules harbour the same variant. By contrast, pathogenic mutations are often transmitted less effectively through the female germline than neutral variants, reflecting selective processes that are only beginning to be understood. Such shifts in heteroplasmy are central to explaining both the rapid evolution of mtDNA and the maintenance of mitochondrial genome integrity. The mitochondrial 'genetic bottleneck' hypothesis, which proposes that only a small proportion of maternal mtDNA molecules are transmitted to offspring, provides one explanation for how variation in heteroplasmy levels arises and thereby creates the substrate on which selection can act.

The observation of major shifts in heteroplasmy level in Holstein cows[51], in which a new heteroplasmic mtDNA variant became homoplasmic within a few generations, first led to the formulation of the genetic bottleneck hypothesis. During oocyte development, restriction of the number of mtDNA molecules that contribute to the next generation results in strong statistical sampling, causing rapid changes in allele frequency (heteroplasmy level) between mother and offspring. Similar segregation patterns have been observed across several species, including humans[40,48] and mice[52]. Measurements of heteroplasmy variance at different stages during germline development revealed a large increase in variance between primordial germ cells (PGCs) and primary oocytes, implicating this developmental stage in the bottleneck mechanism[52] (Fig. 1b). Subsequent work showed a ~1,000-fold reduction in the mtDNA copy number in PGCs compared with oocytes, with the cell mtDNA content falling to ~200 molecules from ~228,000 (ref. 53), with comparable findings in human embryos, bovine systems and zebrafish indicating conservation of this germline mtDNA copy number reduction across species[54–57]. Mathematical modelling using the measured cell mtDNA content and the number of cell divisions (which approximates the duration of the bottleneck) has shown that the measured reduction in mtDNA copy number is sufficient to cause a bottleneck effect that explains the difference in heteroplasmy levels observed in offspring, even in the absence of any selection pressure[53].

The biological significance of this bottleneck remains debated. One possibility is that the marked reduction in mtDNA copy number and resulting shift in heteroplasmy level between germ cells exposes deleterious mtDNA variants to selection, enabling their removal from the germline.

Purifying selection

For neutral genetic variants, the mitochondrial bottleneck effectively accelerates random genetic drift, leading to equal numbers of offspring

Box 1 | Principles of mitochondrial DNA genetics

Mitochondria generate most cellular ATP through tricarboxylic acid cycle and oxidative phosphorylation[1,2], and they contribute to calcium regulation, lipid and amino acid metabolism, and apoptosis[1–4]. These organelles are thought to have originated from a proteobacterium that was engulfed by an archaebacterium around 1.45 billion years ago[137–140]. This endosymbiont was long assumed to fall within the α-proteobacteria, but recent phylogenomic analyses place it on a lineage that branched off before the divergence of sampled α-proteobacteria, rather than within them[141,142]. Following the endosymbiotic event, the genomes of proteobacteria lost most of their genetic material, retaining only a few key genes encoding proteins involved in oxidative phosphorylation, thereby forming the current small double-stranded mitochondrial genome[140]. This mitochondrial DNA (mtDNA) is a circular, double-stranded, intron-less molecule of 16,569 bp that uses a slightly different genetic code than nuclear DNA[143,144]. Located within the mitochondrial matrix, in mammals, mtDNA encodes 13 oxidative phosphorylation proteins, 22 tRNAs and two ribosomal RNAs (rRNAs)[143–145]. Most cells contain multiple copies of mtDNA, varying from dozens in sperm to hundreds of thousands in a mature oocyte[146–148] .

mtDNA replication is a continuous and active process that is independent of the cell cycle[149]. The widely accepted model of mtDNA replication consists of strand displacement: the sole mitochondrial DNA polymerase γ (Polγ) initiates replication at the origin of heavy-strand replication (OriH) to synthesize the heavy strand until it reaches the light-strand replication origin (OriL), triggering synthesis of the light strand. Unidirectional synthesis of the two strands continues until they form a double-stranded mtDNA molecule[150–152]. Despite the high intrinsic proofreading activity of Polγ[153,154] ongoing replication inevitably introduces errors[155], leading to the accumulation of mtDNA mutations with a characteristic mutational signature[124–128]. Pathogenic mutations in protein-coding genes can directly impair the respiratory chain, whereas mutations in tRNA and rRNA genes disrupt the translation of mitochondrial proteins. Limited mtDNA repair mechanisms[156], a lack of protective nucleosome structure[157] and high levels of reactive oxygen species generated within the mitochondrial matrix contribute to mtDNA damage[158–160], although the extent of reactive oxygen species involvement is lower than previously thought[126]. Evolutionary studies consistently show that mtDNA has accumulated nucleotide base substitutions around 10–20 times faster than nuclear genes[48,49].

Across human populations, over 24,000 distinct mtDNA variants have been identified, 130 of which are confirmed to cause disease, according to the database MITOMAP[161]. A new mutation initially involves only a single DNA molecule and remains undetectable until it accumulates to a higher level (for example, through selection or random drift). The coexistence of mutant and wild-type mtDNA within a cell is known as heteroplasmy[51,162]. Cellular dysfunction typically arises when the proportion of mutant molecules, that is, the mutation load, exceeds a critical threshold of between 60% and 80% (ref. 163) (Fig. 1a). The same mutation can present with different clinical symptoms at different heteroplasmy levels (for example, heteroplasmy-associated severity in m.8993T>G (ref. 164)). Random drift often leads to the loss of mtDNA mutations[165,166], but occasionally less severe or neutral mtDNA mutations segregate to high levels, involving all the molecules in a cell — a condition termed homoplasmy. Homoplasmy can be required for clinical phenotypes to manifest, such as in Leber hereditary optic neuropathy[167].

In most animals, except for some fungi and mussels, mtDNA is exclusively maternally transmitted[10–12]. Although it was thought that mtDNA could be biparentally inherited in some humans[168,169], later studies showed that the reported paternal leakage was likely due to contamination by nuclear–mitochondrial DNA segments[170–173]. Beyond a passive dilution of paternal mtDNA in the zygote, an active process of degradation has been observed across species[174], including humans[175], mice[110], *Drosophila melanogaster*[176] and *Caenorhabditis elegans*[177]. Uniparental inheritance contributes to avoiding heteroplasmy-related adverse effects, both genetically and physically[178], as well as to minimizing the spread of selfish genomes[11,179].

with higher or lower heteroplasmy levels than the mother. However, non-neutral mutations behave differently. Early studies of deleterious mtDNA mutations revealed a skewed distribution of heteroplasmy in the offspring compared with the mother, consistent with selection shaping transmission in addition to the bottleneck[36,37,40] (Fig. 1a). Selection can be positive, increasing the transmission of particular mtDNA variants[58,59], or purifying, reducing the transmission of potentially deleterious mutations to offspring.

Evidence for purifying selection

Purifying selection has been observed across many species, including human pedigrees, mouse models of mitochondrial disease and even in non-mammalian species with substantially different germline biology, such as *D. melanogaster*[36,48,60] (Table 1). In mice, purifying selection was first observed in a large 4,696 bp deletion model encompassing loss of six tRNA and seven protein-coding genes in mtDNA[37]. The deletion heteroplasmy declined across successive litters from the same dam, eventually producing offspring free of the deletion. Similarly, a mitochondrial disease model with spontaneous substitution mutations due to a defective proofreading polymerase confirmed that inheritance of non-synonymous mtDNA mutations was lower than expected[36]. Lower-than-expected transmission was also observed in mice with a frameshift mutation in *mt-Nd6* (m.13885insC) but not in those with a non-synonymous substitution mutation in *mt-CoxI* (m.6589T>C), consistent with the latter variant being less deleterious[38]. Subsequent work reported signatures of purifying selection for different classes of mutation in distinct mtDNA genes[61,62].

The data in humans are less consistent. Some mutations show strong negative selection, which explains why large-scale deletions are rarely inherited, whereas some protein-coding mutations do not seem to be subjected to the same selection pressures as in mice[63]. One large UK population study of mother–offspring pairs observed substantial differences between the inheritance of non-synonymous and synonymous mtDNA mutations, with mutations in ribosomal RNA genes being selected against, whereas mutations in the non-coding D-loop, located at the replication origin of the heavy strand (Box 1), were either neutrally transmitted or showed evidence of positive selection[42].

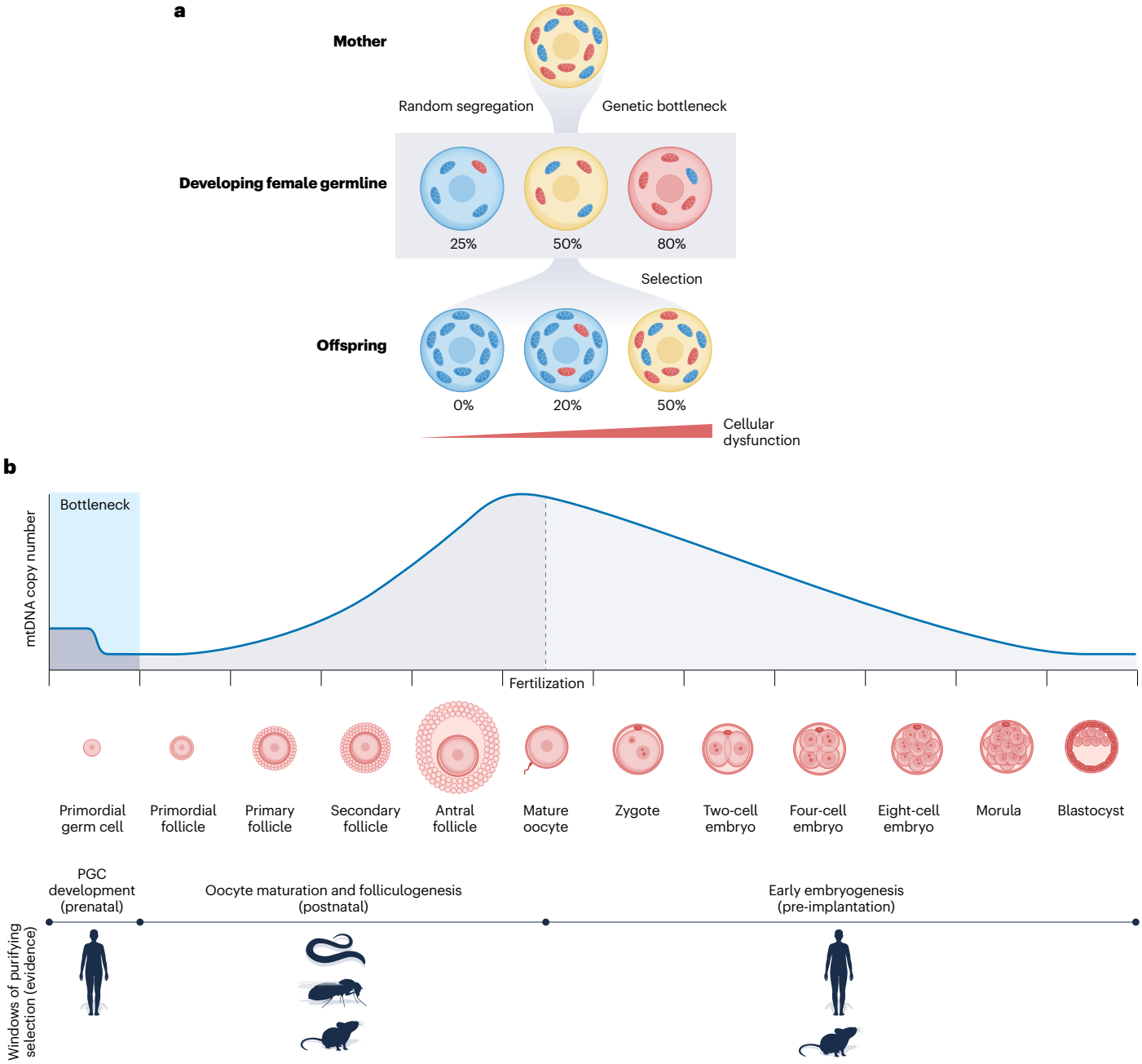


Fig. 1 | Genetic forces shape the inheritance of the mitochondrial genome. **a**, Random segregation, the genetic bottleneck and selection act synergistically to shape mitochondrial DNA (mtDNA) inheritance. Random segregation and the mitochondrial genetic bottleneck generate heteroplasmy variation during germline development. Purifying or positive selection then acts on this variation, altering the heteroplasmy levels transmitted to offspring. Blue and red mitochondria represent alternative mtDNA genotypes. Higher heteroplasmy levels (percentages) are associated with an increased risk of cellular dysfunction. **b**, During primordial germ cell (PGC) development, mtDNA copy number is reduced dramatically, generating the mitochondrial genetic bottleneck. Subsequent mtDNA amplification during folliculogenesis restores high copy numbers in the mature oocyte, whereas mtDNA replication is largely quiescent during pre-implantation development, causing a progressive reduction in mtDNA copy number per cell during cleavage-stage divisions. The bottleneck is thought to increase heteroplasmy variance among germ cells, thereby creating the substrate on which purifying selection can act. Symbols indicate developmental stages for which evidence of purifying selection has been reported in different species.

Table 1 | Reported mechanisms of purifying selection

Involved nuclear gene	Species	mtDNA mutation or model	Selection timing	Outcome	Refs.
Organism level					
NA	Mouse	Homozygous mutant *Polg*	After implantation	Dysfunctional mitochondria impair epigenetic reprogramming and embryo development	79
NA	Mouse	m.5019G>A (*mt-Ta*)	NR	Smaller litter and higher neonatal mortality in a heteroplasmy-dependent manner	80
NA	Mouse	m.5024C>T (*mt-Ta*) m.13885insC (*mt-Nd6*) m.3177G>A (*mt-Nd1*) m.2820G>A (*mt-Nd1*)	NR	No effect on litter size	38,61, 62,80
NA	Mouse	*mt-Cox3* (m.6726A>G; 9138G>A; 9278A>T; 12853C>T; 13441C>T)	After implantation	Embryonic death before E9.5	81
Cellular level					
NA	Mouse	Heterozygous and homozygous mutant *Polg*	Oogenesis	Increase in atretic follicles, fewer mature oocytes and a smaller litter size, thought to be mediated through a lowered NADH/NAD^+ redox state	45
NA	Mouse	m.3177G>A (*mt-Nd1*)	Oogenesis	Higher rate of follicular atresia and lower oocyte maturation rate	85
NA	*Caenorhabditis. elegans*	*uaDf5* mutation (3.1 kb deletion of *nd1, nd2, atp6, ctb-1* and seven tRNA genes)	Oogenesis	Inhibiting programmed cell death in germ cells by mutating the gene encoding the CED-3 caspase led to higher mutation load in progeny but rescued litter size	86
NA	*Drosophila melanogaster*	$mt{:}Col^{T300I}$	Oogenesis	No obvious cell death in germarium cells	60
p35	*D. melanogaster*	$mt{:}Col^{ts}$+$mt{:}nd2^{del1}$	Oogenesis	Overexpression to inhibit cell death failed to weaken selection strength	32
NA	Mouse	*mt-Rnr1* and *mt-Rnr2*	Early embryogenesis	Mutant cells eliminated by cell competition	91
Organelle level					
AMPK	Mouse	De novo mutations and shared variants	Mature oocytes	KO promotes the replication of mutated mtDNA	95
Bcl2l13	Mouse	m.5024C>T	NR	KO hindered the decrease of heteroplasmy in pups	106
BNIP3	*D. melanogaster*	$mt{:}Col^{ts}$+$mt{:}nd2^{del1}$	Cyst	KD inhibited selection during oocyte development by lowering WT mtDNA copies instead of eliminating mutant copies	32
Drp1	*D. melanogaster*	$mt{:}Col^{ts}$+$mt{:}nd2^{del1}$; $mt{:}Col^{T300I}$	Region 2A	OE promoted selection whereas KD inhibited selection by inhibiting mito-fission	32,33
Fis1	*D. melanogaster*	$mt{:}Col^{ts}$+$mt{:}nd2^{del1}$; $mt{:}Col^{T300I}$	Region 2A	KD inhibited selection by inhibiting mito-fission	32,33
FUNDC1	*C. elegans*	uaDf5 (3 kb deletion)	Oocyte–zygote transition	KO inhibited selection by inhibiting programmed mitophagy	107
Marf or *fzo-1* (orthologues of *Mfn1* and *Mfn2*);	*D. melanogaster*	$mt{:}Col^{ts}$+$mt{:}nd2^{del1}$; $mt{:}Col^{T300I}$	Region 2A	OE inhibited selection whereas KD enhanced selection by promoting mito-fission	32
	C. elegans	uaDf5 (3-kb deletion) bguDf1 (1-kb deletion) bguDf2 (4-kb deletion)	Worm development	KO and KD enhanced selection by promoting mito-fission	31
Mitophagy-related genes	Mouse	m.2820G>A	NR	No differential expression of mitophagy-related genes, nor differences in lysosome counts, between WT and mutant oocytes	61
Parkin	Mouse	m.5024C>T	NR	No effect of KO on selection	107[a]
	D. melanogaster	$mt{:}Col^{T300I}$ and $mt{:}ND2^{del1}$	NR	No effect of KD and KO on selection	35
	D. melanogaster	$mt{:}Col^{T300I}$ and $mt{:}ND2^{del1}$	NR	KD blocked selection	102
	C. elegans	uaDf5 (3 kb deletion) bguDf1 (1 kb deletion) bguDf2 (4 kb deletion)	Worm development	KD and KO of *pdr-1* (orthologue) increases heteroplasmy in progeny	104,105

Table 1 (continued) | Reported mechanisms of purifying selection

Involved nuclear gene	Species	mtDNA mutation or model	Selection timing	Outcome	Refs.
Organelle level (continued)					
PINK1	*D. melanogaster*	*mt:CoI*^{T300I}	Oogenesis	PINK1 accumulated on the membrane of mutant mitochondria and inhibited local protein translation, thereby suppressing the replication of mutant mtDNA and ultimately shifting heteroplasmy	101
Polg	*D. melanogaster*	*mt:yak/mt:ND2del1+CoIT300I*	NR	KD promoted selection	34
	D. melanogaster	*mt:CoI*^{T300I} and *mt:ND2*^{del1}	NR	KD promoted selection	102
SOD2	Mouse	De novo mutations and shared variants	Mature oocytes	KO had no effect on the mtDNA mutation pattern	98
Tfam	Mouse	m.5024C>T	NR	KD promoted selection and reduced mutation load in offspring	103
Top1, Top3a, Polg2 and *DNAlig3*	*D. melanogaster*	*mt:CoI*^{T300I} and *mt:ND2*^{del1}	NR	KD enhanced selection	102
Twnk, Ssbp1, Tfam and *Top3a*	*D. melanogaster*	*mt:yak/mt:ND2del1+CoIT300I*	NR	No effect on selection	34
ULK1	Mouse	m.5024C>T	NR	KO inhibited the decrease of heteroplasmy in pups	106
	D. melanogaster	*mt:CoI*^{ts}*+mt:nd2*^{del1}	Cyst	KD inhibited selection during oocyte development by lowering WT mtDNA copies instead of eliminating mutant copies	32
USP30	Mouse	m.5024C>T	Maternal–zygotic transition	KO activated selection against pathogenic mutations	65
mtDNA level					
NA	Human *D. melanogaster*	Large deletion variants	NA	Smaller genome replicates faster	117–119
NA	Human *D. melanogaster*	Mutation affecting affinity for replicative enzymes	NA	Specific mutation-induced increased affinity for replicative enzymes	120,121

E9.5, embryonic day 9.5; KD, knockdown; KO, knockout; mitophagy, mitochondrial autophagy; mito-fission, mitochondrial fission; mtDNA, mitochondrial DNA; NA, not applicable; NR, not reported; OE, over-expression; WT, wild type.

Timing of selection

The studies described above established that selection occurs during maternal mtDNA transmission, but the precise timing remained unclear. A key step towards understanding the underlying biological mechanisms was to determine when selection occurs during germline or early embryo development. Analysis of PGCs and primary oocytes isolated from a mouse transmitting the m.3875delC *mt-Tm* mutation showed that selection occurred after these developmental stages in this model[64] (Fig. 1b). Subsequently, optimized methods enabled the isolation of intermediate follicle stages in three mouse models of clinically relevant non-synonymous mtDNA substitution mutations affecting either tRNA or complex I genes[30,61,62]. In the first study, single germ cells were isolated from mice transmitting the m.5024C>T mutation in the *mt-Ta* gene. This study found that the mtDNA mutation was transmitted preferentially when heteroplasmy levels were low, whereas purifying selection became apparent at high heteroplasmy levels in oocytes[30]. A similar inheritance pattern was reported for women transmitting tRNA mutations such as m.8344A>G, m.8993T>G/C and m.3442A>G, consistent with positive selection acting at low heteroplasmy levels and purifying selection acting at high heteroplasmy levels[30]. A study involving mutations in *mt-Nd1* (m.2820G>A) and *mt-Nd5* (m.12918G>A) expanded this view, revealing strong negative selection during follicle maturation after the primordial follicle stage[61]. Another study focused on the *mt-Nd1* mutation m.3177G>A, with oocytes isolated from all follicle stages, including primordial, primary, secondary and antral. This study showed a significant decrease in heteroplasmy across the secondary follicle group, highlighting this stage as key for the purification of this mutation[62]. Finally, analysis of the m.5019A>G *mt-Ta* mutation identified a different stage for selection, at the start of embryogenesis, through a mitochondrial autophagy (mitophagy)-dependent mechanism[65], and studies of mixed-strain polymorphic heteroplasmies implied two separate waves of subcellular selection, one in oocytes and the other during early embryogenesis[66]. Taken together, these findings indicate that mtDNA selection in mammals is more complex than previously anticipated and suggests the presence of mutation-specific mechanisms that act at different stages during germ cell maturation and early embryonic development. Nevertheless, it should be noted that it remains unclear to what extent these discrepancies reflect technical differences between studies.

Similar patterns of selection against mtDNA mutations during germline development have been observed in non-mammalian species including *D. melanogaster* and *C. elegans* across different mutation classes[60,67], albeit with distinctive features that, at least in part, reflect inter-species differences in germ cell development. Germline purification of a temperature-sensitive *mt-Co1* mutation m.2369C>T (T300I) in *D. melanogaster* showed selection happening during early oogenesis, at the 16-cell cyst/2b late germarium stage (Box 2), which was linked with mtDNA segregation occurring in a similar window[35,68], potentially exposing deleterious variants to selection. Evidence

for selection shortly after the bottleneck has also been observed in humans, in whom non-synonymous variants are selectively depleted during PGC proliferation[55]. However, no further selection during the later stages of folliculogenesis, as seen in mice, has been observed in *D. melanogaster*[61,62], supporting the idea that some mechanisms of purifying selection may be unique to mammals.

Box 2 | Germline transmission across species

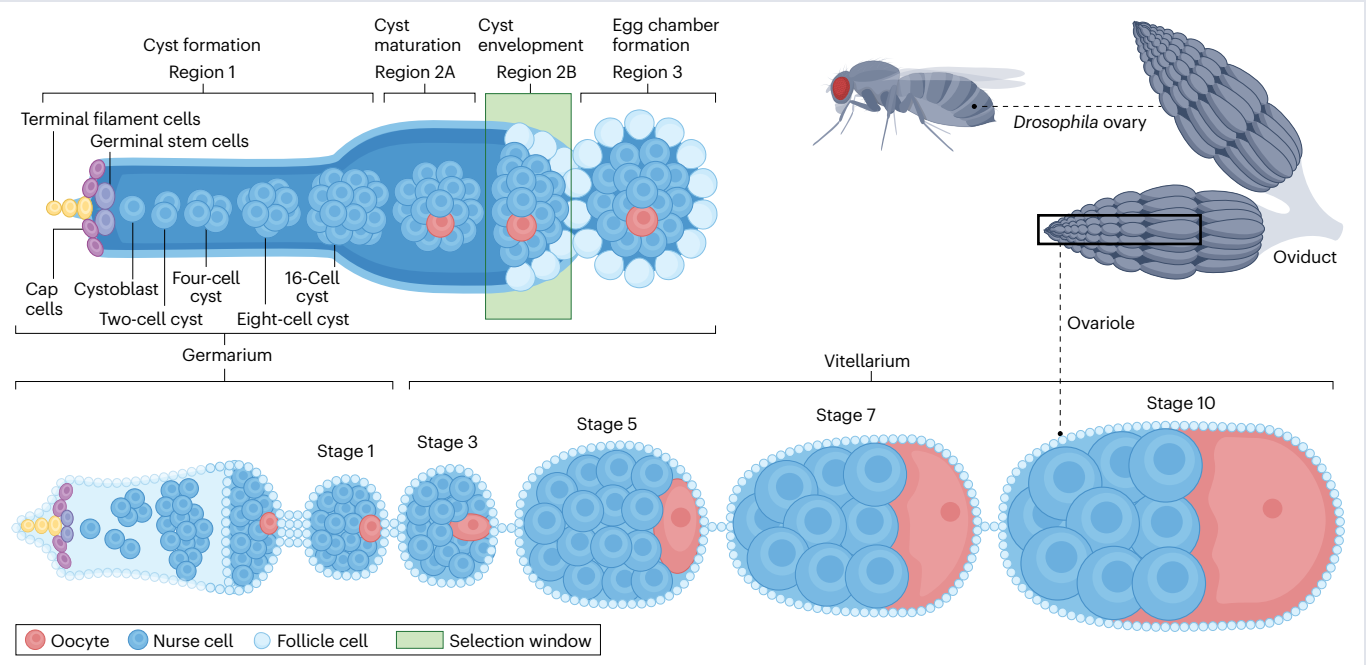
Both mammalian and *Drosophila* germline development begin during embryogenesis, when the primordial germ cells (PGCs) are specified and migrate to the developing gonads, which go on to form the ovaries[180]. Mammalian females harbour two ovaries, subdivided into multiple regions, with the cortex holding the maturing ovarian follicles[181]. The two *Drosophila* ovaries are far more complex structures, each comprising 16–20 ovarioles, which are further divided into the germarium, where the initial cyst maturation takes place, and the vitellarium, for the later stages of development[182] (see the figure).

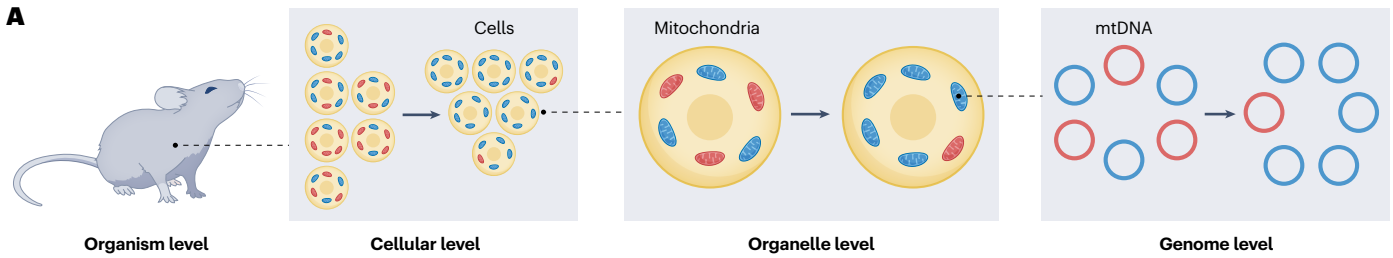
In mammals, PGCs mature into primary oocytes during the embryonic stage (Fig. 1b). At birth, these oocytes are surrounded by supporting granulosa cells and are enclosed within primordial follicles, which constitute a finite ovarian reserve available for ovulation and fertilization throughout life[183]. By contrast, in *Drosophila*, germline stem cells are established long after embryogenesis, at the larval to pupal transition[184]. Unlike mammalian primordial follicles, germline stem cells self-renew throughout adult life, continually generating new oocytes across the lifespan.

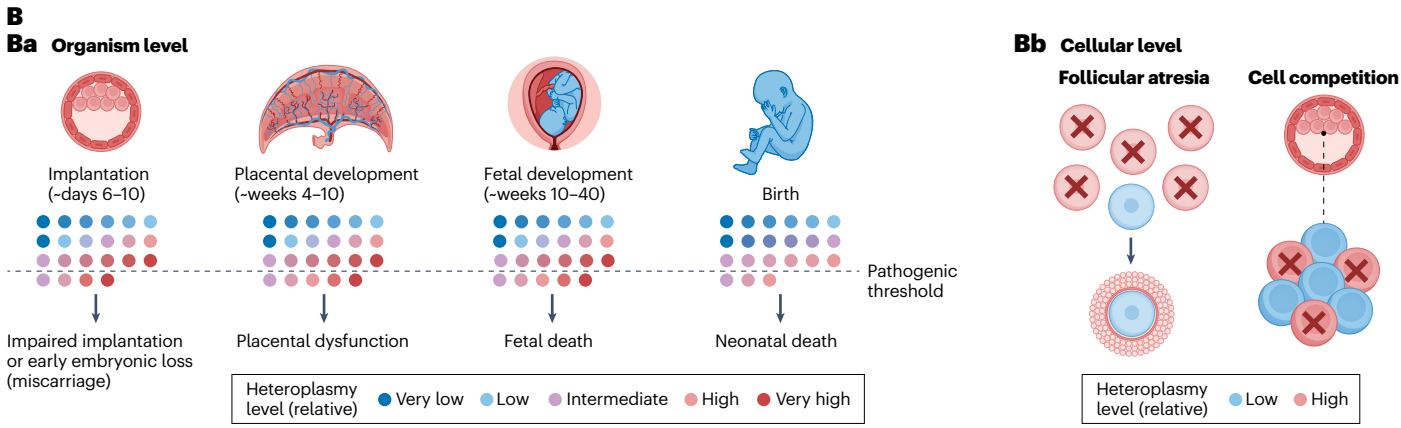
Another major difference is the timing of oocyte specification. Mammalian follicle development begins with an already specified oocyte. In *Drosophila*, germline stem cells first divide to form a cystoblast, which undergoes four rounds of incomplete division into a 16-cell cyst (see the figure). At this point, one cell is specified as the oocyte, whereas the remaining cells adopt supporting nurse-cell fates[182].

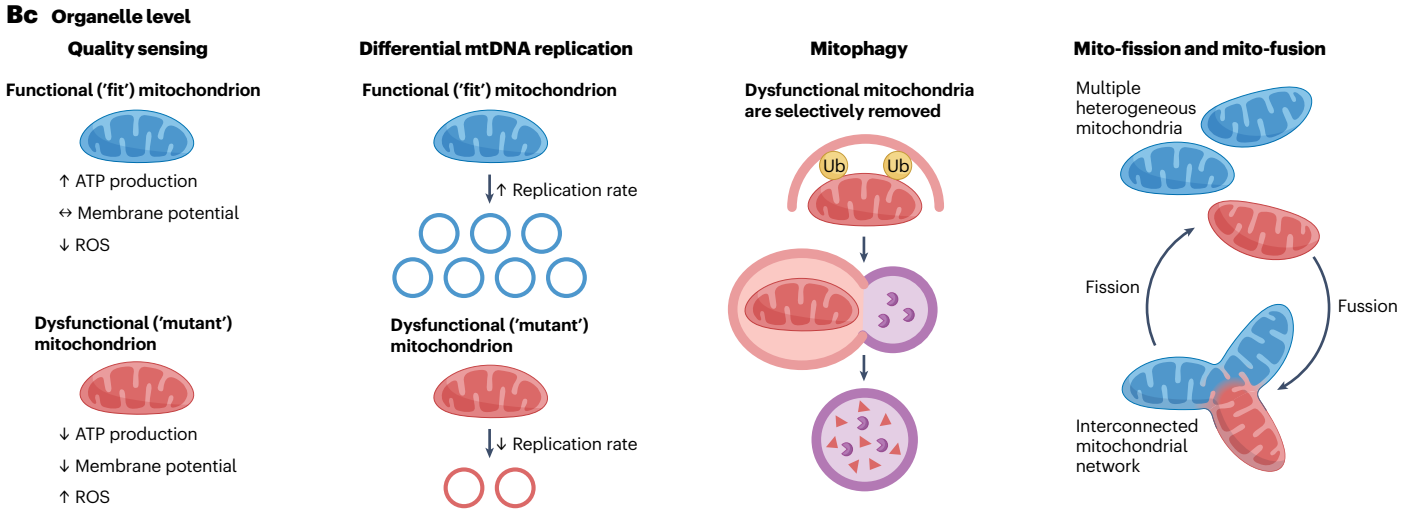
Despite these developmental differences, key features of mitochondrial inheritance are broadly shared across Mammalia and *Drosophila*, including the mitochondrial bottleneck and the broad principles of purifying selection. In mammalian embryogenesis, the mitochondrial bottleneck is associated with a marked reduction in mtDNA copy number during PGC development[52]. In *Drosophila*, an analogous increase in mtDNA segregation occurs during cyst development and oocyte specification in the germarium of the adult lifestage[33]. In both systems, these processes increase heteroplasmy variance and provide an opportunity for selection to act.

Current evidence suggests that the timing of purifying selection differs substantially between *Drosophila* and mammals. In *Drosophila*, selection is observed primarily during oocyte specification in the germarium, immediately after mtDNA segregation[60]. In mammals, selection has been reported at multiple developmental stages, including PGC development[55], during the early stages of follicle maturation, long after the bottleneck and oocyte specification[61,62], and much later in the early embryo[65] (Fig. 1b). Nevertheless, several mechanisms might be at least partially conserved. Mitochondrial fragmentation, mitochondrial quality control and mtDNA replication have all been implicated in selection in both systems[30,32,55,60,93,94]. Although the precise mechanisms and timing differ, collectively, these studies in mammals and *Drosophila* suggest that selection is linked to mitochondrial function and acts to reduce transmission of deleterious mtDNA variants across generations.




A
Cells
Mitochondria
mtDNA
Organism level
Cellular level
Organelle level
Genome level


B
Ba Organism level
Implantation (~days 6–10)
Placental development (~weeks 4–10)
Fetal development (~weeks 10–40)
Birth
Pathogenic threshold
Impaired implantation or early embryonic loss (miscarriage)
Placental dysfunction
Fetal death
Neonatal death
Heteroplasmy level (relative)
Very low
Low
Intermediate
High
Very high
Bb Cellular level
Follicular atresia
Cell competition
Heteroplasmy level (relative)
Low
High


Bc Organelle level
Quality sensing
Functional ('fit') mitochondrion
↑ ATP production
↔ Membrane potential
↓ ROS
Dysfunctional ('mutant') mitochondrion
↓ ATP production
↓ Membrane potential
↑ ROS
Differential mtDNA replication
Functional ('fit') mitochondrion
↑ Replication rate
Dysfunctional ('mutant') mitochondrion
↓ Replication rate
Mitophagy
Dysfunctional mitochondria are selectively removed
Ub
Ub
Mito-fission and mito-fusion
Multiple heterogeneous mitochondria
Fission
Fussion
Interconnected mitochondrial network

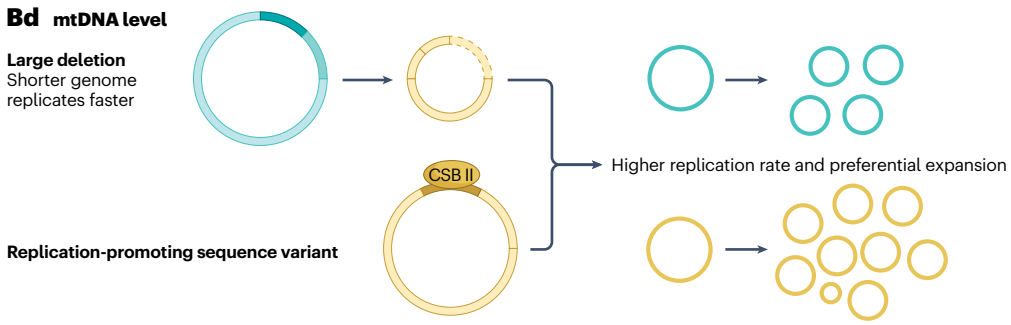

Bd mtDNA level
Large deletion
Shorter genome replicates faster
CSB II
Replication-promoting sequence variant
Higher replication rate and preferential expansion

Fig. 2 | Purifying selection at different biological levels in the mammalian female germline. A, Selection can act at the organism, cellular, organelle or genome (mitochondrial DNA (mtDNA)) level during maternal transmission, with each level potentially contributing to changes in heteroplasmy and the transmission of mtDNA variants. **B**, Representative mechanisms of purifying selection at different biological levels. At the organism level (part **Ba**), high heteroplasmy can impair placental function, fetal development and offspring viability, reducing transmission of deleterious variants. At the cellular level (part **Bb**), germ cells carrying high heteroplasmy can be eliminated through oocyte death, follicular atresia or cell competition after fertilization. At the organelle level (part **Bc**), mitochondrial fitness can influence selection through quality sensing, fitness-dependent mtDNA replication and/or mitochondrial autophagy (mitophagy). Elevated mitochondrial fission (mito-fission) might promote the exposure of mutations to the quality-control system and trigger downstream mitophagy. At the mtDNA level (part **Bd**), intrinsic properties of particular genomes can confer replicative advantages, for example, owing to large-scale deletions or specific variants that confer elevated affinity to replication factors. Mito-fusion, mitochondrial fusion; ROS, reactive oxygen species; Ub, ubiquitin.

Mechanisms of purifying selection

To lower heteroplasmy levels in offspring, in theory, purifying selection could eliminate mutations: (1) at the organismal level, through lethality of embryos or offspring harbouring high levels of heteroplasmy; (2) at the cellular level, through cell competition, follicular atresia or oocyte apoptosis; (3) at the organellar level, through fitness-dependent mtDNA replication or mitophagy; or (4) at the genomic level, through mechanisms such as intrinsic replication advantages of specific mtDNA variants (Fig. 2). Here, we summarize current evidence of selection occurring at all four levels.

Selection at the organism level

Embryo implantation and fetal development are energy-intensive processes. Therefore, mitochondrial dysfunction caused by high-level heteroplasmy could lead to embryonic defects, placental dysfunction and, ultimately, fetal or neonatal death, resulting in lower average heteroplasmy levels in surviving offspring. In keeping with this hypothesis, women carrying mtDNA mutations have an increased risk of obstetric abnormalities[69,70], including prenatal pregnancy loss and neonatal death[69,70]. A study analysing 287 pregnancies indicated that women carrying mtDNA mutations have an approximately 25-fold higher risk of miscarriage and an approximately 35% reduction in live birth rate than women in the control[71]. The variants m.8344A>G in the *MT-TK* gene[72] and m.3243A>G in the *MT-TL1* gene[70,73] have been associated with a higher risk of miscarriage, and analysis of chorionic villus tissue from early miscarriages revealed a higher proportion of embryos carrying mtDNA mutations compared with controls[74]. Several studies showed that m.3243A>G was associated with placental dysfunction and a resulting higher risk of pre-term birth (7–53% in women carrying the mutation versus 5.3% in women not carrying it)[70,73,75–77], which would greatly reduce neonatal survival in the absence of modern medical care[78].

Animal models support the hypothesis that mtDNA mutations impair post-implantation embryonic development and lead to placental dysfunction[79]. Oocytes homozygous for mutant *Polg*, which encodes the catalytic subunit of DNA polymerase γ (Polγ)[19] — the only reported mtDNA polymerase in mammalian mitochondria — carry an approximately twofold higher burden of mtDNA mutations. This mutation load resulted in ~20% fewer live births, lower birthweight and lower placental weight[79]. Further analysis revealed damaged mitochondria, but no nuclear compromise, affecting epigenetic reprogramming and ultimately the developmental potential of embryos[79]. Similarly, mice carrying the m.5019G>A mutation in the *mt-Ta* gene showed smaller litter sizes and higher neonatal mortality, which correlated with increasing heteroplasmy levels[80]. *mt-Cox3* mutations in chimeric blastocysts derived from embryonic stem cells did not affect embryonic implantation, but most embryos died before embryonic day 9.5 (ref. 81). By contrast, data from mouse models of m.5024C>T in *mt-Ta* (ref. 80), m.13885insC in *mt-Nd6* (ref. 38), and m.3177G>A (ref. 62) and m.2820G>A (ref. 61) both in *mt-Nd1*, showed no significant effect on litter size, again suggesting that selection patterns differ depending on the mtDNA mutation. Taken together, these findings suggest that selection could act against high-level pathogenic heteroplasmy and thereby lower the mutation load in the next generation through impaired placental function and/or embryogenesis.

Selection at the cellular level

In the mammalian germline, individual oocytes develop largely independently before fertilization. Consequently, selection can act either on the oocyte itself before fertilization or on embryonic cell lineages after fertilization.

Before fertilization: oocyte death. As mentioned above, the bottleneck effect results in marked differences in heteroplasmy levels among oocytes. A high mtDNA mutation load could impair oocyte development and even cause oocyte death, with only oocytes carrying lower heteroplasmy levels being successfully ovulated. During mammalian germline development, in each menstrual cycle, a cohort of multiple primordial follicles is recruited for growth, but only a small fraction — typically one in humans and about ten in mice — progress to maturity, whereas the others are subjected to programmed cell death. This physiological process is known as follicular dominance[82]. Generally, only 0.02% of oocytes grow and are ultimately ovulated in humans[83].

Inter-species comparisons have shown that a narrower bottleneck correlates with a higher rate of physiological follicular atresia, implicating atresia in selection[84]. mtDNA variants in *Polg*-mutant female mice were associated with an increase in atretic follicles, fewer mature oocytes and a smaller litter size, thought to be mediated through a lowered NADH/NAD^+ redox state, with a reduced litter size rescuable by nicotinamide mononucleotide[45]. Recently, researchers using in vitro follicle culture technology observed that oocytes carrying the m.3177G>A *mt-Nd1* mutation exhibited a much higher rate of follicular atresia and lower oocyte maturation rate, despite similar fertilization and blastocyst formation rates[85]. In *C. elegans*, the *uaDf5* mutation (a large 3.1 kb deletion of protein-encoding genes *nd1*, *nd2*, *atp6* and *ctb-1*, as well as seven tRNAs) impaired litter size and increased embryonic lethality[86]. Inhibiting programmed cell death in germ cells by mutating *ced-3* caspase led to higher mtDNA mutation load in progeny but rescued litter size, indicating selection through oocyte death[86]. By contrast, no obvious cell death occurred in germarium cells in *D. melanogaster* carrying the *mt-Co1* mutation m.2369C>T (T300I)[60], and overexpression of p35 to inhibit cell death failed to weaken selection strength[32].

After fertilization: embryonic cell competition. After fertilization, embryonic cells continue to divide, although mtDNA replication is largely quiescent during pre-implantation stages[87–89]. Consequently,

the pool of mtDNA inherited from the oocyte is partitioned among the dividing blastomeres until the blastocyst stage, resulting in a progressive reduction in mtDNA copy number per cell (Fig. 1b). Random segregation of mtDNA types between the daughter cells can lead to different heteroplasmy levels in different blastomeres[88,89] (Fig. 1a). Cell competition (a process by which cells with lower relative fitness are outcompeted by their fitter neighbours as they divide over time), growth restriction or the death of cells with high mutation load could reduce heteroplasmy in the resulting organism. Indeed, a higher burden of mtDNA variants was found in aberrant human aneuploid embryos, compared with euploid embryos, suggestive of a potential role in developmental failure[90]. Furthermore, embryonic cells carrying higher mtDNA mutation load may be less developmentally competitive than neighbouring cells, thereby reducing overall heteroplasmy in the organism. Indeed, single-cell transcriptional profiling in early embryogenesis showed that cells carrying variants in *mt-Rnr1* and *mt-Rnr2* exhibited mitochondrial dysfunction and were eliminated by cell competition[91].

Selection at the organelle level

Each individual oocyte harbours hundreds of thousands of mitochondria. For selection to act at this level, the oocyte must be able to distinguish functional from mutant mitochondria. At the organelle level, the recognition of mutant genomes is thought to rely on sensing changes in mitochondrial function, including reduced mitochondrial inner membrane potential, impaired production of ATP or increased production of reactive oxygen species. Moreover, downstream mechanisms such as selective mtDNA replication and mitophagy can alter heteroplasmy by preferentially propagating or removing particular mitochondria. Finally, mitochondrial fission can expose functional differences between mitochondria.

Mitochondrial quality sensing. An appropriate mitochondrial membrane potential is indispensable for ATP synthesis and is also a key measure of mitochondrial quality monitored by the cell[92]. A decreased membrane potential triggers the clearance of damaged mitochondria in cultured cells[92]. In general, pathogenic mutations interfering with the electron transport chain lead to a decrease in membrane potential, providing a potential trigger for purifying selection. However, pathogenic mutations in the mtDNA *mt-Atp6* and *mt-Atp8* genes can have the opposite effect, increasing the membrane potential by obstructing proton return through ATP synthase[93,94]. Notably, mutations in the *mt-Atp6* and *mt-Atp8* genes undergo weaker purifying selection or, in some instances, may even be subjected to positive selection[30,36,48,93,94]. In humans, carriers of the m.8993T>G/C mutation in *MT-ATP6* show an increased membrane potential[93,94] and transmit higher heteroplasmy levels to their offspring[30]. The seemingly paradoxical relationship between pathogenicity and strength in *MT-ATP6* and *MT-ATP8* mtDNA mutations suggests that a decreased membrane potential acts as a trigger for purifying selection. However, artificially rescuing the membrane potential of mutant mitochondria in *D. melanogaster* failed to suppress the selection process[95].

Sensing ATP levels could also trigger selection against pathogenic mtDNA mutations. Knockout of the gene encoding the ATP sensor AMPK unexpectedly promoted the replication of mutant mtDNA copies in mature oocytes[95]. Finally, the accumulation of reactive oxygen species has also been proposed to trigger purifying selection through activation of mitochondrial quality-control systems[96,97]. However, increasing endogenous reactive oxygen species through knockout of *Sod2* had no effect on the mtDNA mutation pattern in mature oocytes[98].

Fitness-dependent mtDNA replication. One potential mechanism of purifying selection is the preferential replication of mtDNA in functional versus mutant mitochondria, which could increase the relative abundance of healthy mitochondria over time, thereby decreasing heteroplasmy. Consistent with this hypothesis, disrupting the mitochondrial membrane potential abolished mtDNA replication in *D. melanogaster*, particularly in region 2B of the germarium[60] (Box 2). This finding suggests that mitochondrial fitness is a prerequisite for mtDNA replication during oogenesis. Accordingly, mtDNA replication was markedly reduced at non-permissive temperatures in flies carrying the temperature-sensitive *mt-Co1* mutation m.2369C>T (T300I)[60]. During oogenesis, OXPHOS is activated in germ cells to meet rising energetic demands[55,99,100]. In *Drosophila*, OXPHOS activation in the late germarium is associated with increased expression of mtDNA replication factors, raising the possibility that mtDNA replication contributes to selection[55,99,100]. Similarly, in human PGCs, selection against mtDNA mutations was observed alongside the shift from glycolysis to OXPHOS[55]. Therefore, in both species, selection may occur during developmental transitions associated with increased mitochondrial activity, suggesting that OXPHOS-dependent mitochondrial function has a role in selection. In *Drosophila*, the mechanism underlying shifts in heteroplasmy levels involved mitochondrial protein kinase PINK1, which was found to selectively accumulate on depolarized, mutant mitochondria and inhibit local protein translation in a mitophagy-independent manner[101]. An in situ imaging approach subsequently confirmed that wild-type and mutant mtDNA molecules replicate at different rates[68].

Genome-wide screening showed that the knockdown of the mtDNA polymerase gene *Polg* increased selection against pathogenic mutations in *D. melanogaster*[34]. However, it should be noted that this does not mean that *Polg* is directly involved in the selection mechanism but rather that its knockdown enhances other selection processes, possibly through a bottleneck effect linked to mtDNA copy number. Notably, in this study, the selection effect was specific to the catalytic subunit *Polg*, as reducing the dosage of the essential accessory subunit or other genes related to mtDNA replication, such as *Twnk*, *Ssbp1*, *Tfam* and *Top3a*, did not have the same effect[34]. However, in another large functional screen in *D. melanogaster*, knockdown of *Polg* as well as other replication-related genes, including *Top1*, *Top3a* and *Polg2*, was observed to enhance selection[102]. Knockdown of *Tfam* also promoted selection and reduced mutation load in offspring, possibly by narrowing the bottleneck[103]. Taken together, these studies show that perturbation of the replication machinery modifies selection strength, but the specific molecular mechanisms require further investigation.

Mitophagy. Mitophagy selectively eliminates damaged or dysfunctional mitochondria through lysosomes[92], which could contribute to removing deleterious variants from the germline. Multiple model organism studies have implicated mitophagy-related genes in the selection process. For example, knockdown or knockout of the mitophagy regulator *pdr-1* in *C. elegans*, a homologue of Parkin (*PRKN* gene), increases heteroplasmy levels in progeny[31,104,105]. In *D. melanogaster*, programmed germline mitophagy during oogenesis has been observed immediately before the selection window[68,102] (Box 2); by reducing opportunities for mutant genomes to be functionally complemented by wild-type genomes, mitophagy might strengthen purifying

selection. In female mice carrying the m.5024C>T mutation in the *mt-Ta* gene, knockout of the mitophagy factors *Ulk1* or *Bcl2l13* inhibited the decrease in heteroplasmy typically observed in offspring[106], providing indirect evidence for a role of mitophagy in mtDNA selection. Direct evidence for a role of mitophagy in selection has been reported in oocytes from *C. elegans*, in which mitophagy removes mutant mitochondria during the oocyte-to-zygote transition[107]. Consistent with this finding, in mice, loss or inhibition of USP30, a negative regulator of ubiquitin-mediated mitophagy, during the maternal-to-zygotic transition enhanced selection against pathogenic mtDNA mutations and reduced heteroplasmy levels in offspring[65].

Nevertheless, there may be mitophagy-independent roles for the same gene products in oocytes. For example, in *D. melanogaster*, PINK1 seems to influence mtDNA selection by inhibiting local translation in germ cells rather than by activating Parkin-mediated mitophagy[101], and knocking out *parkin* in *D. melanogaster*[35] or mice[106] had no effect on selection. In a large functional screen, knockdown of Parkin did block selection, but Parkin was shown not to be involved in mitophagy activation[102], which suggests a mitophagy-independent role for Parkin in the selection process. Knockdown of the mitophagy proteins Atg1, a homologue of Ulk1, and Bnip3 in *D. melanogaster* inhibited selection during oocyte development by lowering the number of wild-type mtDNA copies[32]; this finding suggests that these classic mitophagy-related proteins influence selection by promoting the maintenance or propagation of functional mtDNA, rather than through removal of mutant mitochondria. In mice, no differential expression of mitophagy-related genes nor differences in lysosome counts were observed between wild-type and mutant oocytes carrying the m.2820G>A mutation in *mt-Nd1* (ref. 61). In keeping with this finding, independent work suggests that mitophagy can be impaired in mouse oocytes under some conditions[108]. By contrast, mitophagy is highly activated physiologically after fertilization in mice[109,110], suggesting stage-specific regulation.

In conclusion, emerging evidence implicates mitophagy in mtDNA selection, but considerable gaps in knowledge remain, particularly with respect to differences between species and mutation types.

Mitochondrial fission and mitochondrial fusion. Somatic cells usually contain more than one mtDNA molecule per organelle on average[111], which enables wild-type molecules to compensate for mutant mtDNA. However, if mitochondria undergo highly active fission, each organelle will eventually contain only a single mtDNA molecule, potentially exposing mitochondrial functional differences and facilitating downstream selection mechanisms, such as selective replication and mitophagy. Therefore, mitochondrial dynamics can provide the conditions for selection to occur, even if they are not directly involved in the selection mechanism itself.

In *D. melanogaster*, mitochondria in germ cells are smaller, rounder and less networked than those in somatic cells[32,33,112]. Fluorescence labelling experiments revealed barely any matrix exchange between mitochondria in oocytes but high levels of exchange between germline stem cells[32]. Artificially impairing mitochondrial fragmentation by overexpressing *mitofusin (Marf)*, the sole orthologue of both *Mfn1* and *Mfn2* (ref. 32), knocking down *Drp1* (refs. 32,33) or knocking down *Fis1* (ref. 33) significantly inhibited selection, whereas promoting mitochondrial fission by knocking down *Marf* resulted in a less connected mitochondrial network and enhanced selection[32]. Consistent with this finding, compromising mitochondrial fusion by knocking out the *Mfn1/2* orthologue *fzo-1* drove the complete elimination of *uaDf5* mutation within three to four generations in *C. elegans*[31]. In mammalian oocytes, mitochondria are also highly fragmented[113,114], often containing a single mtDNA molecule per organelle[87,115,116]; nevertheless, to our knowledge, direct evidence for a role of fusion or fission in mtDNA selection in mammals has yet to emerge.

Selection at the mtDNA level

Selection at the mtDNA level can be defined as mechanisms in which the mtDNA molecule itself gains a transmission or replication advantage, rather than selection acting on the downstream consequences of a mtDNA mutation. In other words, one mtDNA variant outcompetes another owing to intrinsic properties of the genome, rather than because it produces a 'healthier' mitochondrion.

One example is provided by so-called 'selfish' mtDNA variants, which can increase in frequency because they replicate more efficiently than competing genomes. Such advantages can arise because large deletion variants reduce the size of genomes, which are then copied more rapidly[117–119], or through mutation-induced increased affinity for replicative enzymes[120,121].

Further evidence comes from studies of mitochondrial replacement. Human embryos reconstructed by mitochondrial replacement therapy inevitably retain a small amount (<2%) of the original mtDNA. In some derived stem cell lines, this minority mtDNA population preferentially expanded over subsequent passages and eventually reestablished the original haplotype[121–123]. This phenomenon occurred without detectable differences in OXPHOS function between the competing haplotypes. Instead, the phenomenon was linked to a polymorphism in conserved sequence box II (CSB II) within the mtDNA control region, which suggests that the replication advantage arose from sequence-specific properties of the mtDNA molecule itself[121–123].

Maternal-age effects on selection

As early as 2007, older female mice carrying m.7759_12454del4696 were shown to deliver pups with lower heteroplasmy levels than those born to younger mothers[37]. More recently, two groups independently showed in mice that maternal age is associated with a strong selection effect on point mutations in mitochondrial genes encoding complex I, reducing the mutation load in offspring[61,62]. The m.2820G>A *mt-Nd1* mutation was subjected to strong age-related selection, and the mutation load in progeny approached zero when the mother reached 20 weeks of age at conception[61]. Spatial transcriptome analysis of oocytes from secondary follicles suggested that enhanced mTORC1-regulated protein translation has a role in the selection process. Inhibition of mTORC1 by rapamycin for 49 days significantly weakened the observed selection and thus led to an increased mutation load in mature oocytes[61]. The m.12918G>A *mt-Nd5* mutation exhibited a similar maternal-age-related selection pattern, but the onset of intense selection occurred at an older maternal age (8–9 weeks for m.2820G>A versus 22–23 weeks for m.12918G>A), again pointing towards mutation-specific effects. Finally, studying the m.3177G>A *mt-Nd1* mutation revealed a continuous decrease in heteroplasmy shift with maternal age through the lifespan, with heteroplasmy levels in oocytes approaching zero for mothers reaching 1 year of age[62]. Interestingly, the above-mentioned mutations underwent age-enhanced selection irrespective of the maternal heteroplasmy level (usually measured in an ear or tail biopsy), and litter size was not impaired. This observation suggests that the age effect occurs largely during oogenesis[37,61,62]. In keeping with this interpretation, no age-related selection effect was apparent in other tissues[37,61,62].

The detailed molecular mechanism underpinning age-related selection remains poorly understood and requires further investigation.

One possible explanation is that age-enhanced selection acts as a compensatory response to the increased mutational load and declining mitochondrial function that occurs with age. Imperfect proofreading of Polγ leads to de novo mtDNA mutations arising through ongoing mtDNA replication during life[124–128]. Indeed, many studies have shown that mtDNA mutations accumulate in somatic tissues, potentially contributing to the ageing process[125,129–134]. Human pedigree studies of 39 and 249 mother–baby pairs also found a positive correlation between de novo mtDNA mutations in offspring and maternal age[40,135]. However, in human oocytes, the mutation loads were observed to remain stable or even increased with ageing[45,129,136], albeit always at lower levels than in somatic cells[125,129–134]. Duplex sequencing showed that ageing is associated with increased de novo mutations in blood and saliva but not in human oocytes[129]. In macaque[130] and mice[50], the mutation frequency was increased in oocytes of older individuals compared with younger ones. However, the frequency was still much lower than in liver[130], heart[130], brain[50] or skeletal muscle[50,130], possibly because there is less mtDNA replication in oocytes, or because selection against de novo mutations suppresses their accumulation in oocytes. In summary, age-enhanced purifying selection, observed in multiple models, could serve to resist age-related de novo mtDNA mutation and maintain mtDNA integrity in the germline.

Conclusions and future perspectives

The inheritance of mtDNA is actively shaped by competing genetic forces that act on heteroplasmy variation generated by the germline genetic bottleneck. Positive selection seems to occur under specific circumstances but is much less understood than purifying selection, which can occur at different stages of transmission through powerful and multi-layered biological mechanisms, with differences observed between species and within species for different mtDNA mutations.

Although considerable progress has been made in recent years in identifying the precise stage of selection for several different classes of mtDNA mutation, including the m.3177G>A mutation[62] in *mt-Nd1* and the m.5024C>T mutation in *mt-Ta* (ref. 30), we are only beginning to understand the immense complexity of highly context-dependent purifying selection. Elucidating the details is the next great challenge and will require comprehensive analyses across female germline development and multiple mutational contexts to fully define interdependencies.

Clarifying why the same mutation undergoes opposite selection patterns in different contexts and exactly how the oocyte 'senses' mutations and then determines selection pattern and strength will help to elucidate how species adapt to environmental changes and how selection shapes evolutionary trajectories. Single-cell studies have shown that even low levels of heteroplasmy can induce a transcriptional response[80], indicating the presence of mechanisms that detect mtDNA variation below the traditional threshold for biochemical expression. Nevertheless, open questions remain. What determines the level at which a mutation is selected? Does selection at different levels interact? How does the nuclear genome influence selection, and how has this changed throughout evolution? These compelling questions undoubtedly merit concerted research efforts. Exploring the interplay between the nuclear and mitochondrial genomes will help to delineate the relationship between selection and energy adaptation, as well as open avenues for clinical therapies.

Future work will need to integrate potential mechanisms into a coherent model that explains how selection is initiated, executed and regulated at the molecular level. Understanding how purifying selection acts on the transmission of mtDNA mutations paves the way for developing interventions to prevent mtDNA-related diseases, both rare single-gene disorders and common diseases in which inherited low-level heteroplasmy might contribute to late-onset disease risk. Different mechanisms may act for or against a specific mtDNA variant at different times, which potentially explains why a mutation with severe cellular effects can be propagated in the germline. The combined forces acting on selection likely explain the landscape of genetic variants seen at the population level and influence which variants cause recurrent rare diseases (for example, why m.3243A>G is the most common pathogenic mtDNA mutation in humans). How to prevent the transmission of pathogenic mtDNA mutations remains unclear, or whether it is possible to expand knowledge in the germline to somatic tissues to prevent or treat mitochondrial disease. Emerging techniques, such as large-scale single-cell analyses, conditional knockout models and single-cell CRISPR screens, promise to provide a holistic view of mitochondrial genetics. Uncovering the mechanisms of this mysterious but powerful organelle will likely yield substantial benefits to humans.

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Author contributions

Q.X. and H.E.W. researched the literature. All authors contributed substantially to discussions of the content, wrote the article, and reviewed and/or edited the manuscript before submission.

Competing interests

M.M. is a co-founder, shareholder and member of the Scientific Advisory Board of Pretzel Therapeutics, Inc. The other authors declare no competing interests.

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